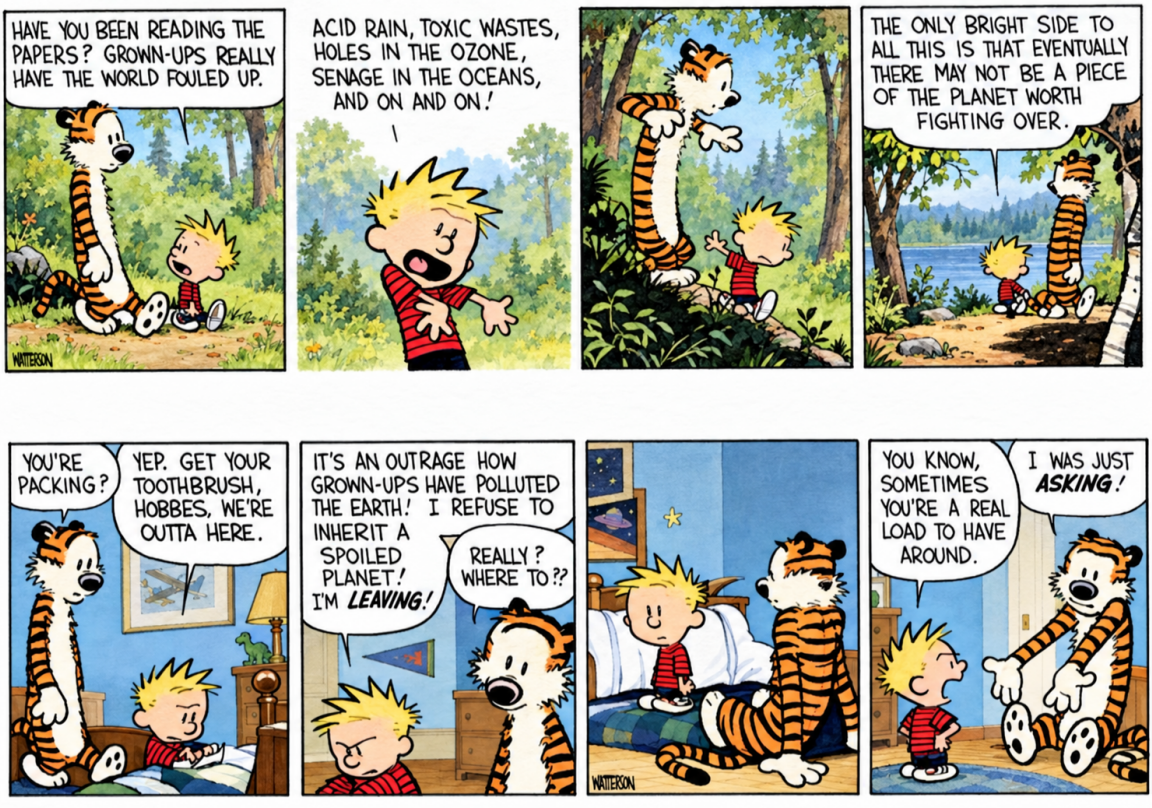
Session - 12

AWS Sagemaker 1

Oct 19, 2026



Agenda

- What is Sagemaker
- How to train & deploy ML models

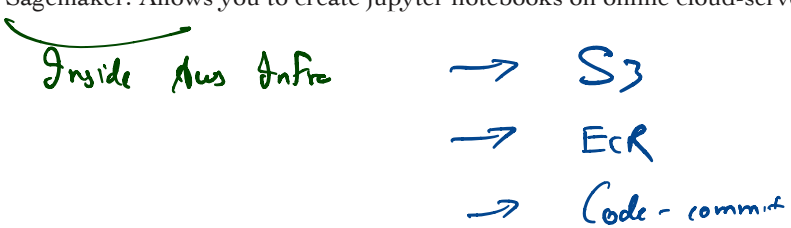
Different ways of Deploying ML models?

1. Streamlit: When I want to show-case my ML models, as POC/MVP.
2. Flask/Fast Api + NGROK: For Exposing API to my comrades.
3. ECS: ECS Fargate + ALB is production level deployment.
4. EC2: EC2 + ALB is also production level deployment.
5. Sagemaker

Why do we need SageMaker??

You are not allowed to download or copy even a fraction of data on any local computer.

Sagemaker: Allows you to create jupyter notebooks on online cloud-server.



Google GCP -> Vertex AI
Azure -> Azure ML
Databricks -> Databricks.

Running instance (much cheaper): Used Primarily for EDA.

Training Instance (relatively expensive): Used only for Training

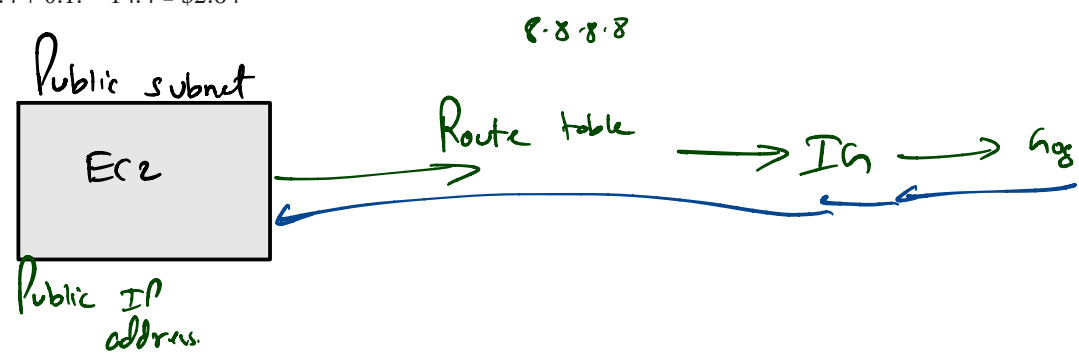
1. p2.16xlarge - $14.4

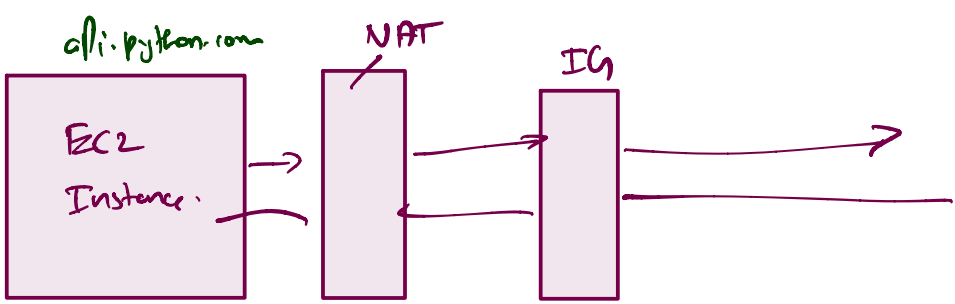
3 hours on EDA/Exploration - 14.4 * 3
0.1 hour for training: 1.44

$44.64

2. m5.2xlarge for EDA: $0.4

3 * 0.4 + 0.1. * 14.4 = $2.64





Carrier GRADE NAT — ISPs

8.8.8.8 : 1000

8.8.8.8 :16000



watch.

IP – address for m.

200. 192. 32. 216

HTTP: 80

HTTPS: 443

AWS networking: how the five pieces fit

Milepost API · one VPC · two public subnets · shared routing and security rules

Example traffic path

Route-table association

Your browser / internet
Call either instance's public IPv4 address
HTTP · TCP port 8000

Direct IPv4 access requires
Gateway + route + public IP
+ security rules that allow the connection

Internet Gateway
Attached to the VPC
Maps public ↔ private IPv4 addresses

This is a logical network view.
The gateway serves the VPC;
it is not placed inside one subnet.

VPC · 10.0.0.0/16

milepost-ec2-vpc · us-west-2
Your logically isolated network

Call A or B using its public IP

Public subnet A

10.0.1.0/24
One address range · one AZ: us-west-2a

Security group: sg-api
Applied to this instance's network interface

EC2 A · Milepost API on TCP 8000
Private IPv4: 10.0.1.10
Public IPv4: PUBLIC_IP_A

IN TCP 8000 from YOUR_PUBLIC_IP/32
OUT All IPv4 traffic to 0.0.0.0/0

Public subnet B

10.0.2.0/24
One address range · one AZ: us-west-2b

Security group: sg-api
Applied to this instance's network interface

EC2 B · Milepost API on TCP 8000
Private IPv4: 10.0.2.10
Public IPv4: PUBLIC_IP_B

IN TCP 8000 from YOUR_PUBLIC_IP/32
OUT All IPv4 traffic to 0.0.0.0/0

Both subnets use this table

Shared route table · rt-public — VPC resource

DESTINATION	TARGET	MEANING
10.0.0.0/16	local	Within the VPC
0.0.0.0/0	Internet Gateway	Default route

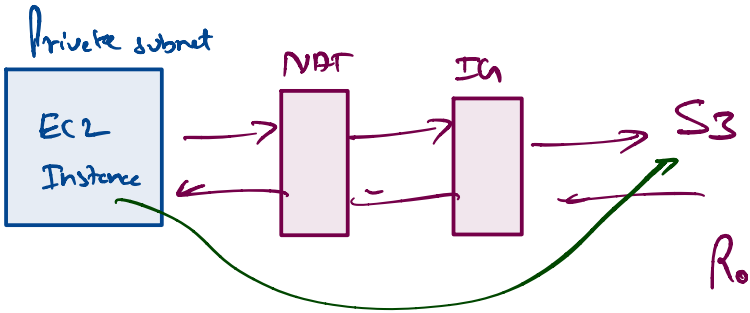
Each subnet uses one route table at a time; multiple subnets can share it.

Public ≠ open to everyone
The IGW route makes a subnet public. Direct internet access also needs a public IPv4 address and matching security rules on the instance.

Routes choose; rules permit
The local route provides a path between subnets. It does not grant permission. Sharing a security group does not automatically allow A and B to talk.

Responses are remembered
Security groups are stateful: replies to allowed connections are permitted automatically. Other unsolicited inbound traffic is blocked in this example.

Scope: IPv4. API running; other network and host settings assumed permissive. PUBLIC_IP_A/B are placeholders.
Based on AWS VPC and EC2 documentation · Paths are schematic; dashed gray lines represent configuration, not traffic.



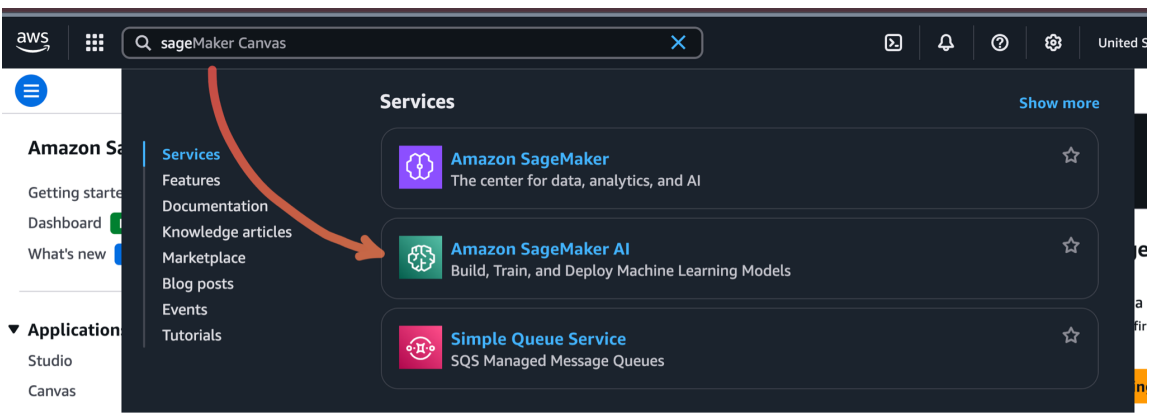
VPC Endpoints gateway.

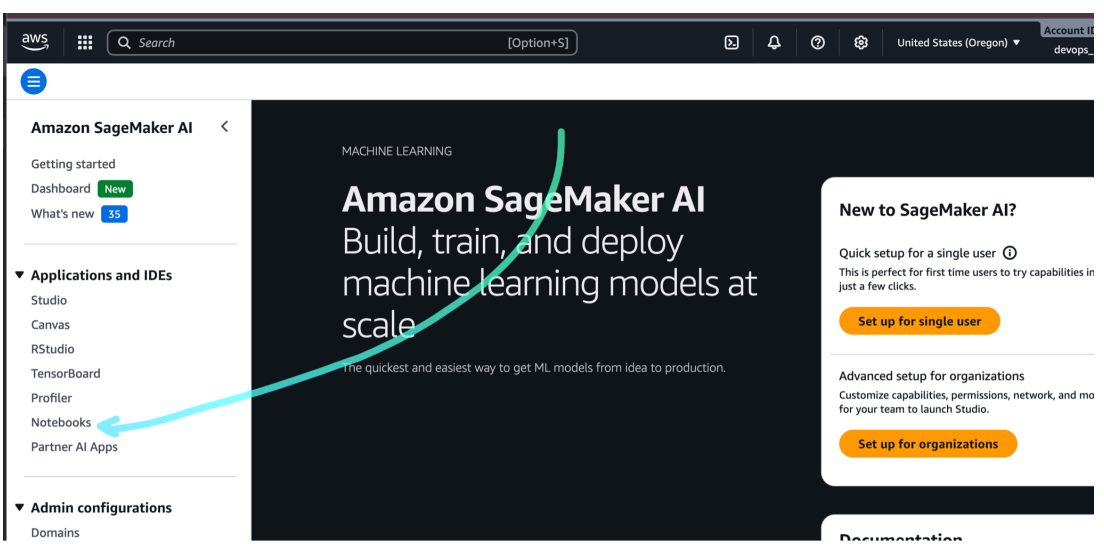
Route table

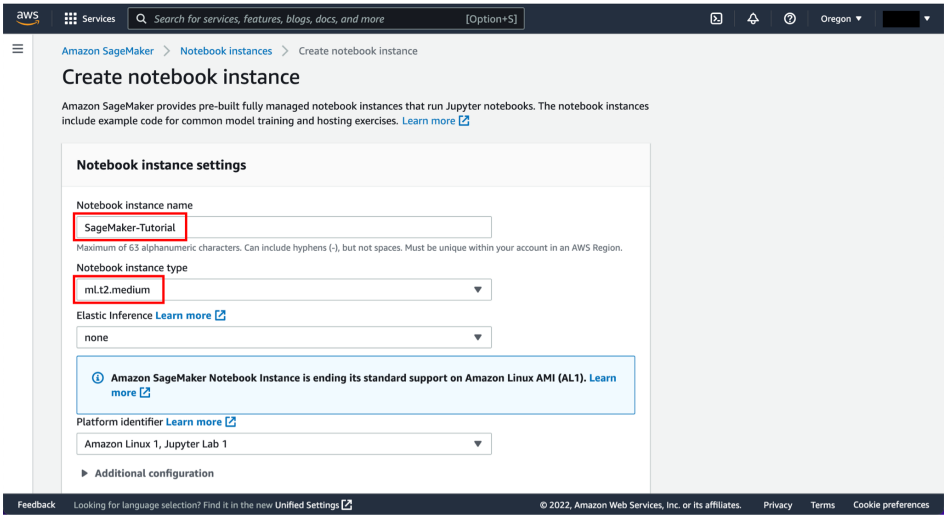
10.0.1.0 → Local

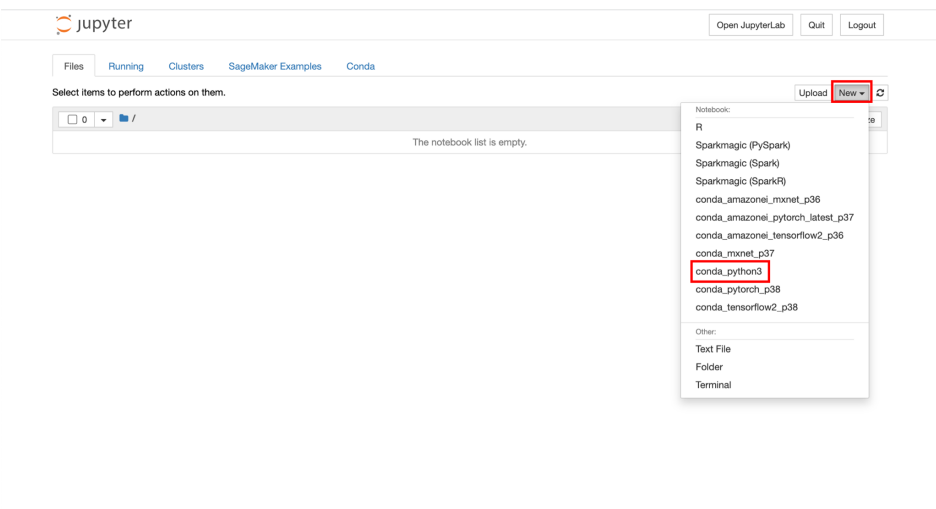
0.0.0.0/0 → NAT

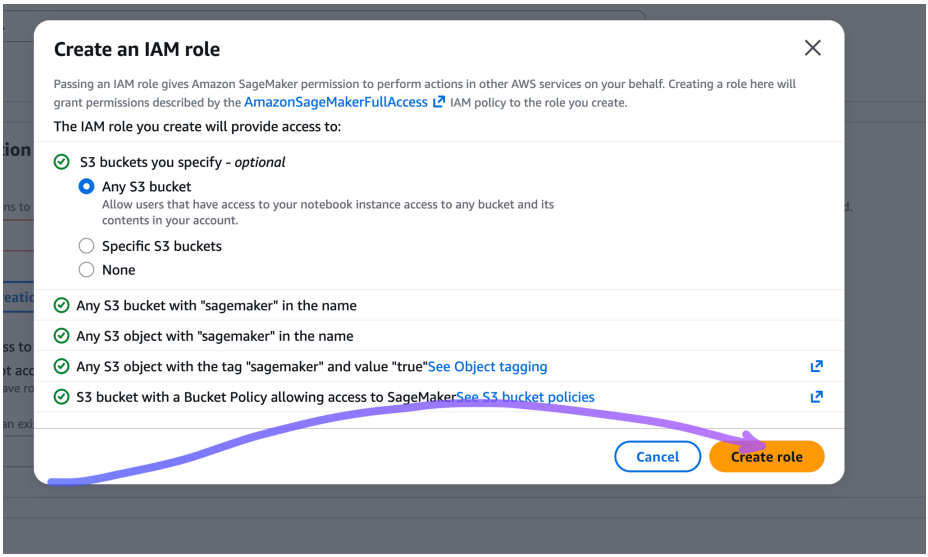
S3-^^* → S3


aws
sageMaker Canvas
United S
Amazon Sa
Getting starte
Dashboard
What's new
Application
Studio
Canvas
Services
Features
Documentation
Knowledge articles
Marketplace
Blog posts
Events
Tutorials
Services
Show more
Amazon SageMaker
The center for data, analytics, and AI
Amazon SageMaker AI
Build, Train, and Deploy Machine Learning Models
Simple Queue Service
SQS Managed Message Queues


aws
Search
[Option+S]
United States (Oregon)
Amazon SageMaker AI
Getting started
Dashboard
New
What's new
35
Applications and IDEs
Studio
Canvas
RStudio
TensorBoard
Profiler
Notebooks
Partner AI Apps
Admin configurations
Domains
MACHINE LEARNING
Amazon SageMaker AI
Build, train, and deploy machine learning models at scale
The quickest and easiest way to get ML models from idea to production.
New to SageMaker AI?
Quick setup for a single user
This is perfect for first time users to try capabilities in just a few clicks.
Set up for single user
Advanced setup for organizations
Customize capabilities, permissions, network, and mo for your team to launch Studio.
Set up for organizations


Services
Search for services, features, blogs, docs, and more [Option+S]
Oregon
Amazon SageMaker > Notebook instances > Create notebook instance
Create notebook instance
Amazon SageMaker provides pre-built fully managed notebook instances that run Jupyter notebooks. The notebook instances include example code for common model training and hosting exercises. Learn more
Notebook instance settings
Notebook instance name
SageMaker-Tutorial
Maximum of 63 alphanumeric characters. Can include hyphens (-), but not spaces. Must be unique within your account in an AWS Region.
Notebook instance type
ml.t2.medium
Elastic Inference Learn more
none
Amazon SageMaker Notebook Instance is ending its standard support on Amazon Linux AMI (AL1). Learn more
Platform identifier Learn more
Amazon Linux 1, Jupyter Lab 1
Additional configuration
Feedback
Looking for language selection? Find it in the new Unified Settings
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jupyter
Open JupyterLab
Quit
Logout
Files
Running
Clusters
SageMaker Examples
Conda
Select items to perform actions on them.
Upload
New
The notebook list is empty.
Notebook:
R
Sparkmagic (PySpark)
Sparkmagic (Spark)
Sparkmagic (SparkR)
conda_amazonei_mxnet_p36
conda_amazonei_pytorch_latest_p37
conda_amazonei_tensorflow2_p36
conda_mxnet_p37
conda_python3
conda_pytorch_p38
conda_tensorflow2_p38
Other:
Text File
Folder
Terminal



Digressing to S3 now, for creating bucket

